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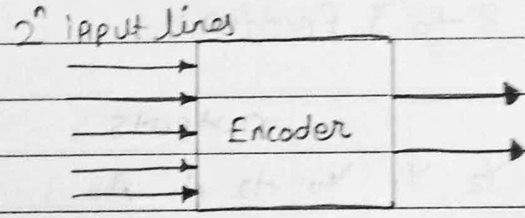
course/program :- BTech CSE SY
Subject :- DLM

DLM assignment 2.

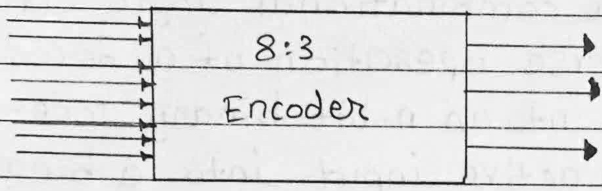
Q.1) Explain Encoder in detail with the help of one example
Sol^n :- An encoder is a combinational logic circuit that perform the inverse operation of a decoder. It converts 2^n input lines into an n-bit binary code - essentially at "encodes" the active input into a binary input.



How an Encoder works?
- A typical encoder has:
  - Input : 2^n input lines, each representing a unique Signal or event.
  - output : n output lines, representing the binary code corresponding to the active input.
- for example an 8-to-3 encoder has:
  - 8 input lines (D_0 to D_7)
  - 3 output lines (Y_2, Y_1, Y_0)

Octal to Binary Encoder (8-to-3 Encoder)

- The 8 to 3 Encoder or octal to binary encoder consists of 8 inputs: Y_7 to Y_0 and 3 outputs A_2, A_1 & A_0. Each input line corresponds to each octal digit and three outputs generates corresponding binary code. The figure below shows the logical symbol of octal to the binary encoder.



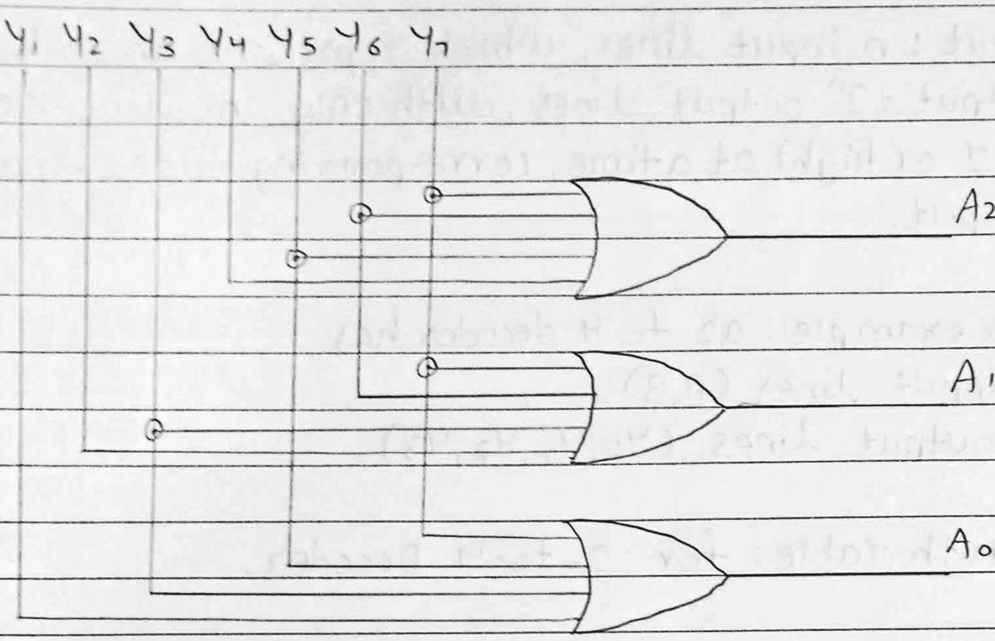
· Truth Table for 8 to 3 Encoder:

Inputs								Outputs		
Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Logical expression for A_2, A_1 and A_0

$A_2 = Y_7 + Y_6 + Y_5 + Y_4$
$A_1 = Y_7 + Y_6 + Y_3 + Y_2$
$A_0 = Y_7 + V_5 + Y_3 + Y_1$

The above two boolean functions A_2, A_1 and A_0 can be implement using four input OR gates.



Example use case.
Encoders are commonly used in

- Data compression:
- Reducing the number of bits required to represent information.
- Keyboards:- where each key corresponds to a unique input and the encoder generates a binary code that identifies which key was passed.
- Multiplexers:- To select the desired data input line.

Q.2) Explain decoder in detail with the help of one example.

Sol^n: A decoder is a combinational logic circuit that converts binary data from n input lines into a maximum of 2^n unique output lines essentially, it "decodes" the binary input into a specific active output lines.

How a decoder works?
- A typical decoder has:

- Input: n input lines, which represent an n-bit binary no.
- output: 2^n output lines, with only one line active (set to 1 or high) at a time, corresponding to the binary value of input.

- for example: a 2 to 4 decoder has
- 2 input lines (A,B)
- 4 output lines (Y_0, Y_1, Y_2, Y_3)

Truth Table: for 2-to-4 Decoder.

Input	output
A	B
0	0
0	1
1	0
1	1

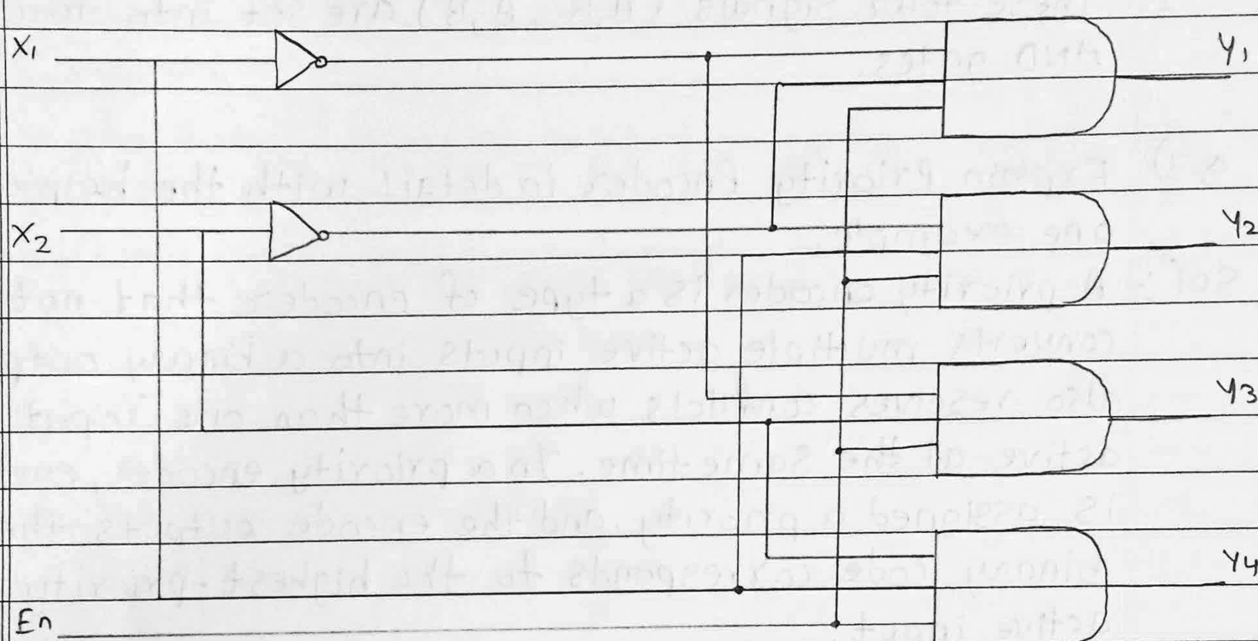
En	X_2	X_1	y_4	y_3	y_2	y_1
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

- Explaination of truth table:
- when A=0 and B=0, the binary value is 00 The decoder activates the output Y_0, which corresponds to 1000.
- when A=0 and B=1, the binary value is 01. The decoder activates the output Y_1, which corresponds to 01.
- when A=1 and B=0, the binary value is 10 The decoder activates the output Y_2, which corresponds to 10.
- When A=1 and B=1, the binary value is 11. The decoder activats the output Y_3, which corresponds to 11.

Circuit diagram for a 2-to-4 Decoder.

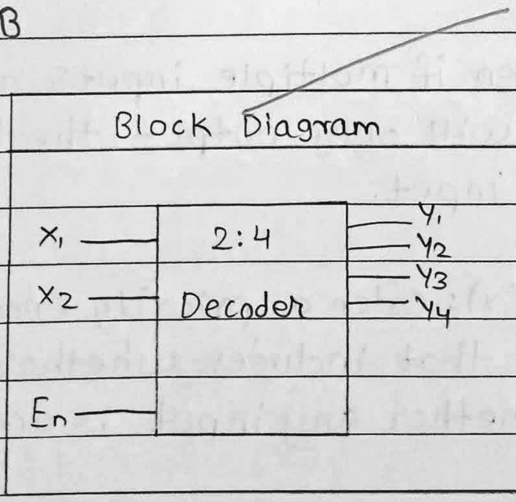
- The circuit consists of:

1. NOT Gates: To generate the component of each input.
2. AND Gates: To combine the inputs and their components to generate the correct output.



Logic Equation for output:

- $Y_0 = \bar{A} \cdot \bar{B}$
- $Y_1 = \bar{A} \cdot B$
- $Y_2 = A \cdot \bar{B}$
- $Y_3 = A \cdot B$



Each AND gate outputs a HIGH signal (1) for its corresponding combination of inputs, which all other outputs remains LOW (0).

1. Input A & B go into NOT gates to generate $\bar{A}$ & $\bar{B}$
2. These four signals (A, B, $\bar{A}$, $\bar{B}$) are set into four AND gates.

Q.3) Explain Priority Encoder in detail with the help of one example.

Sol^n:- A priority encoder is a type of encoder that not only converts multiple active inputs into a binary output but also resolves conflicts when more than one input is active at the same time. In a priority encoder, each input is assigned a priority, and the encode outputs the binary code corresponds to the highest-priority active input.

Key characteristics of a priority Encode:

1. Priority Assignment: Each input has a predetermined priority level. The input with the highest priority will be selected if multiple inputs are active.

2. Single output: Even if multiple inputs are active the priority encoder will only output the binary code of the highest priority input.

3. Output Validity (V): Often a priority encoder includes an additional output that includes whether the output is valid (i.e., whether any input is active).

4 to 2 Priority Encoder.

This is also referred to as 4-bit priority which consists of 4 inputs and 2 outputs lines since an encoder contains 2^n input lines and n output lines. The third output is 'V', which is considered as a valid but indicator and it is set to 1 when more than one input line is high or active (1).

If the valid bit is equal to '0' then all the inputs are '0' In this case, the other 2 output lines are considered as don't case conditions denoted by x.

The truth table of a 4 to 2 priority encoder is shown below.

D_3	D_2	D_1	D_0	A	B	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

from above truth table we can observe that D_3, D_2, D_1, D_0 are the inputs, A and B are the output and V is the valid bit indicator. Here D_3 input is the highest priority input and D_0 is the lowest priority input.

when input D_3 is active high (1), which has the highest priority irrespective of all other input lines, then the output of the 4-bit priority encoder is 11.

when the D3 input is active low and the D2 is active high that has the next highest priority irrespective of all other input lines, then the output is BA = 10.

when D3, D2 outputs are active low, and the D1 is active high and has the next highest priority regardless of the remaining input line the output will be BA = 01.

the output expression can be determined for a 4-bit encode with the help of a knaught map (K-map) as shown below.

D1D2 \ D3D4		01	11	10
00	x	0	0	0
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$B = D_2 + D_3$

D1D2 \ D3D4	00	01	11	10
00	x	0	1	1
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$A = D_3 + D_1 D_2$

$A = D_3 + D_1 D_2'$

$B = D_2 + D_3$

$V = D_0 + D_1 + D_2 + D_3$

~~4 to 2 priority encoder circuit diagram.~~